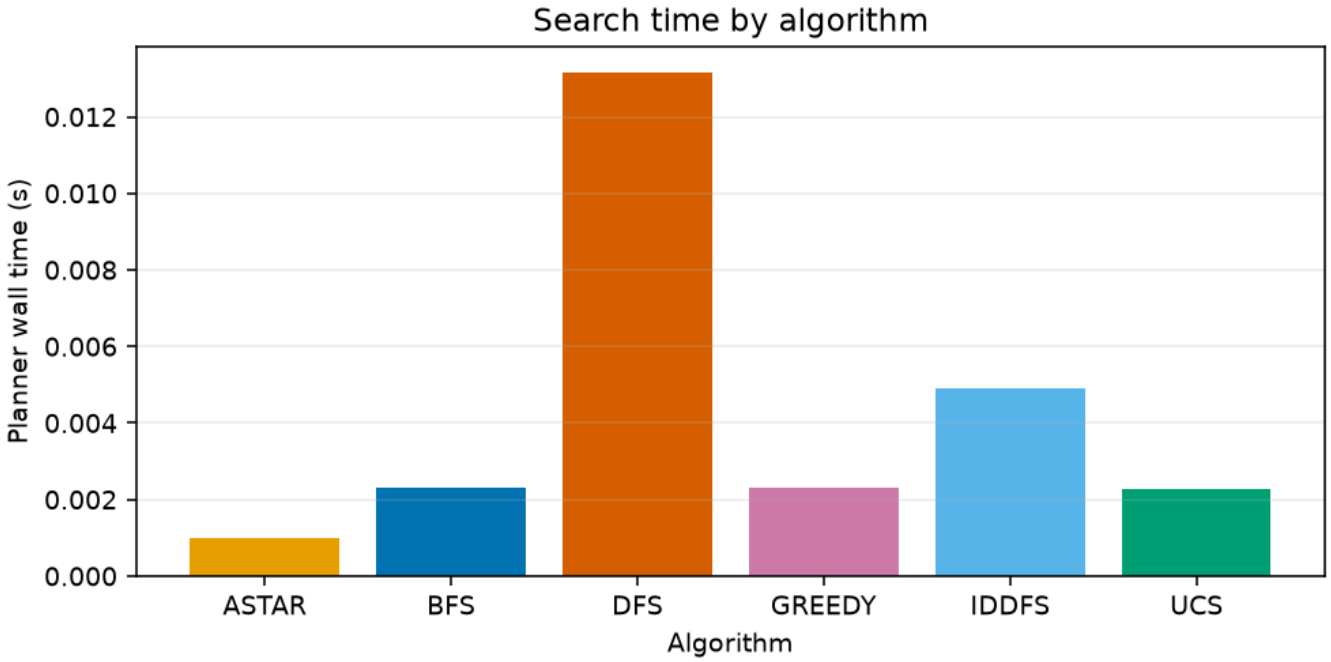
Coverage comparison

Runs share map, robot, policy, neighbor order, cost model, heuristic, and mission mode.

Algorithm	Expanded	Search (s)	Steps	Cost	Energy	Coverage %	Stop
ASTAR	34	0.0009993999265134335	34	34.0	2.31999999999978	100.0	mission_complete
BFS	72	0.0022909999277681112	34	34.0	2.31999999999978	100.0	mission_complete
DFS	279	0.013165500364266336	56	56.0	3.4199999999997175	100.0	mission_complete
GREEDY	34	0.0022912001004442573	34	34.0	2.31999999999978	100.0	mission_complete
IDDFS	49	0.004917699960060418	34	34.0	2.31999999999978	100.0	mission_complete
UCS	72	0.0022714996011927724	34	34.0	2.31999999999978	100.0	mission_complete



Assumptions and limitations

Map grid: 6 × 6; resolution 1.0 m/cell. Movement: four-neighbor-cardinal; neighbor order: ['up', 'right', 'down', 'left']; heuristic: manhattan; cost model: {'transition': 'unit'}; tie-breaking: recorded algorithm default; coverage policy: {'id': 'nearest_uncleaned_manhattan_v1', 'configuration': {}}.

A single scenario does not support a generalized superiority claim. DFS depends on neighbor order. Greedy search is not guaranteed to return an optimal path. IDDFS may repeat expansions across depth limits. Simulated battery values are abstract model outputs, not real appliance measurements.